

COURSE TITLE : ADHESIVE TECHNOLOGY
COURSE CODE : 6121
COURSE CATEGORY : A
PERIODS PER WEEK : 5
SEMESTER : 6
PERIODS PER SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
I	Introduction to Adhesive	3
	Principles of Gluing	10
	Test	1
II	Natural Glues	9
	Composition of Adhesives	9
	Test	1
III	Synthetic resins	18
	Preparation of Resin	8
	Test	1
IV	Phenolic resins	6
	Latest developments	8
	Test	1
	Total	75

Course Outcome :

Module	G.O	Student will be able to:
1	1	understand general considerations on glues and gluing
	2	comprehend the test on adhesive
2	1	understand composition of adhesive
	2	understand the application of resin
3	1	understand the synthetic resins
	2	comprehend the production of synthetic resins
4	1	understand the manufacture of phenolic resins and rf resins
	2	understand epoxy resins

Specific outcome

MODULE-I

1.1.0 Understand General Considerations on Glues and Gluing

- 1.1.1 Define adhesive, adherents, adhesion, glue
- 1.1.2 Explain adhesive, adherents, adhesion, glue with examples
- 1.1.3 Describe the general consideration of wood glues
- 1.1.4 Describe cohesion and adhesion
- 1.1.5 Describe intermolecular forces, polarity, primary and secondary forces (valencies)
- 1.1.6 Describe the influence of temperature, concentration and amount of hardening of glue joints
- 1.1.7 Describe PH value on the hardening of glue joints
- 1.1.8 Describe Rheology of glues
- 1.1.9 Describe phenomena of hardening

1.2.0 Comprehend the Test on Adhesive

- 1.2.1 Describe the effect of wood species and moisture content on the strength of glue joint
- 1.2.2 Describe the strength improvement and stresses in glue joints
- 1.2.3 Describe the pre – treatments of wood prior to gluing
- 1.2.4 Describe glue spreading
- 1.2.5 Describe defects in gluing
- 1.2.6 Identify the test on adhesive
- 1.2.7 Describe pH, viscosity, water tolerance, solid content, specific gravity, solubility
- 1.2.8 Describe estimation of phenol, formaldehyde

MODULE-II

2.1.0 Understand Composition of Adhesive

- 2.1.1 State Composition of Adhesive Discuss the additives used in glue mixing such as back bone polymer, solvent, thinner, catalyst, hardner, extender, filler, fertilizer, plasticiser, preservatives
- 2.1.2 Explain the process by which adhesive develop strength
- 2.1.3 Describe glue mixing type of mixers
- 2.1.4 Describe viscosity shelf life and pot life of resin
- 2.1.5 Explain addition of caustic soda or acid to control PH during mixing

2.2.0 Understand the Application of Resin

- 2.2.1 List Classification of adhesives
- 2.2.2 Describe the classification of glues
- 2.2.3 Explain natural glues – Animal glues, Vegetable glues, casein glues, Blood albumin glues, starches, dextrans, Natural gums
- 2.2.4 Describe shellac, Asphalt, natural & synthetic rubber
- 2.2.5 Explain glues on the basis of cellulose-cellulose acetates, cellulose esters and silicate of soda

MODULE-III

3.1.0 Understand the Synthetic Resins

- 3.1.1 Describe – synthetic resin, thermosetting resin
- 3.1.2 Discuss the development of synthetic resins
- 3.1.3 Discuss amino resins – urea formaldehyde and melamine formaldehyde resin
- 3.1.4 Discuss the method of manufacture of U.F resins
- 3.1.5 Describe chemistry of urea resins – mono methylol urea, dimethylol urea, trimethylol urea.
- 3.1.6 Explain the properties and uses of UF resins
- 3.1.7 Explain the Lab preparation of UF resin
- 3.1.8 Explain the manufacture of urea moulding powder and important variables and control in the manufacture of moulding powder
- 3.1.9 Describe the manufacture of UMF resin
- 3.1.10 Describe Melamine formaldehyde resin
- 3.1.11 Describe the method of manufacture of MF resin

3.2.0 Comprehend the Production of Synthetic Resins

- 3.2.1 Describe the properties of MF resin
- 3.2.2 Describe the Lab preparation of MF resin
- 3.2.3 Describe the moulding powder and powdered MF resin
- 3.2.4 Describe laminating resins and surface coating
- 3.2.5 Describe industrial application of MF resins & UF resin
- 3.2.6 Explain about resin kettle
- 3.2.7 Illustrate resin kettle and mark its important parts.
- 3.2.8 Describe the chemistry of RF resin
- 3.2.9 Describe the method of manufacture of RF resin
- 3.2.10 Describe the industrial application of RF resin
- 3.2.11 Describe the laboratory preparation of RF resin

MODULE-IV

4.1.0 Understand the manufacture of phenolic resins and rf resins.

- 4.1.1 State PF and RF resins
- 4.1.2 Describe phenolic resins
- 4.1.3 Describe the chemistry of PF resin – resole and Novolak resins
- 4.1.4 Explain the method of manufacture of PF resin
- 4.1.5 Describe the lab preparation of PF resin
- 4.1.6 Describe the properties of glue joints with PF resins
- 4.1.7 Describe the industrial applications of PF resin
- 4.1.8 Describe the manufacture of resorcinol phenol formaldehyde resin
- 4.1.9 Describe adhesives based on cashew nut shell liquid (CNSL)

4.2.0 Understand Epoxy Resins

- 4.2.1** State epoxy resins
- 4.2.2** Describe the method of manufacture of epoxy resin
- 4.2.3** Describe the thermoplastic resin
- 4.2.4** Describe the cellulose adhesives & polyvinyl acetate
- 4.2.5** State the polymerization
- 4.2.6** Describe the polyethylene, polyamides polyesters and vinyl acetate
- 4.2.7** State the properties and uses of epoxy resin
- 4.2.8** Describe polyurethane adhesives (isocyanate resins)
- 4.2.9** Describe the raw materials used in polyurethane adhesives
- 4.2.10** Describe the properties and uses of polyurethane adhesives
- 4.2.11** Describe the advantages of polyurethane adhesives

CONTENT DETAILS

MODULE 1

General considerations about gluing – adhesion, adhesive, adhered- Physical and chemical properties of gluing – cohesion and adhesion, Intermolecular forces- polarity, primary and secondary valencies, Rheology of glue, Influence of pH value on the hardening of glue joints physical properties of adhesive film- thickness of adhesive film- pre-treatments of wood prior to gluing. Glue spreading Defects in gluing – effect of wood species and moisture content on the strength of glue joint – strength improvement and stresses in glue joints. 3. Test on adhesive- PH, Viscosity, water tolerance, solid content by oven dry method, specific gravity, solubility, estimation of phenol and formaldehyde. Determination of and value of resin. Test results of various glue joints]

MODULE II

Composition of adhesive – introduction – back bone polymer – solvent and thinner – catalyst – hardner – extension – filler – fortifier – plasticiser – preservative process by which adhesive develop strength –n glue mixing, types of mixers. Addition of caustic soda or acid to control the PH during mixing. Viscosity shelf life and pot life of resin . Natural glues Introduction – animal glues, vegetable glues, starch glues, soyabean glues, protein glues, casein glues, blood albumin glues, starch glues, soyabean glues, protein glues casein glues, blood albumin glues starches, dextrans, natural gums.Shellac, asphalt, natural and synthetic rubber. Glues on the basis of cellulose – cellulose acetates, cellulose esters and silicate of soda.

MODULE- III

Introduction to synthetic resin, thermosetting resin . Amino resins – UF resin & MF resin manufacturing procedure and chemistry of UF and M.F resins. Lab preparation of UF & MF resins. Manufacture of UMF resin. Urea moulding powder, melamine moulding powder industrial applications of UF & MF resins . Resin kettle – diagram Chemistry and method of manufacture of RF resin. Industrial application of RF resin.

MODULE IV

Introduction to phenolic resins – chemistry and method of manufacture of PF resin – lab preparation of PF resin. Properties of glue joints with PF resin – phenolic moulding powder – Industrial application of PF resin .Chemistry and method of manufacture of RF resin. Industrial application of RF resin, CNSL – based adhesives and its by products.. Thermoplastic resins – introduction, Polymerization acetate cellulose adhesives polymerization, polyethylene, polyamides polysters and vinyl acetate. .Epoxy resin and its raw materials method of manufacture of epoxy resin properties and user of epoxy resin, polyurethane adhesives – raw materials – properties and uses. Advantage of polyurethane adhesives over urea and phenolic resins

REFERENCE

1. F.P Kollmann, Principles of wood science & Technology ,Springer
2. Synthetic resins & their industrial applications – SBP – Board of consultants & Engineers New Delhi
3. Lecture notes on plywood technology – IPIRI Bangalore 4. Adhesion &